

Testing the palaeoclimate extrapolation limit of a hybrid machine-learning atmospheric model

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Abstract

Hybrid climate models that combine a dynamical core with learned physics now reach operational weather skill at a fraction of the cost of conventional general circulation models, yet their behaviour outside the training climate remains largely untested. Here the hybrid model NeuralGCM, trained on reanalysis of 2001–2018, is evaluated along a ladder of six Holocene time slices between the present and 6 ka before present, in which orbital forcing and ocean boundary conditions depart progressively from the modern state. More than one hundred independent 14-month simulations are performed, with sea surface conditions derived from an 8300-year transient simulation of the coupled model AWIESM2, which also serves as the verification target, so that all skill scores measure fidelity to a single plausible Holocene rather than to a reconstruction. A present-day control reproduces ERA5 with a global temperature bias below 0.4 K and a pattern correlation of 0.997. Along the Holocene ladder no abrupt loss of skill is found. Instead the model splits into components that age differently. The monsoon circulation response strengthens monotonically with the insolation anomaly all the way to 6 ka, and the winter temperature response is captured increasingly well as the forced signal rises above internal variability. The summer response over land remains poor at every slice, and the learned precipitation scheme reverses the sign of the West African monsoon response even though the simulated moisture convergence agrees with the target model. Our results reveal that, across the components tested here, physics carried by equations extrapolates into past climates while physics carried by fitted weights does not, a boundary that defines which palaeoclimate applications of such models are presently defensible.

1 Introduction

Machine-learning models of the atmosphere have progressed within a few years from proof of concept to operational competitiveness. Purely data-driven forecast systems now match or exceed leading numerical weather prediction at medium range (Pathak et al., 2022; Bi et al., 2023; Lam et al., 2023), and hybrid designs couple a differentiable dynamical core to learned parameterisations so that the resolved flow remains governed by the primitive equations (Kochkov et al., 2024). The hybrid model NeuralGCM simulates weather and climate at a small fraction of the computational cost of a conventional atmospheric model, and multi-decadal AMIP-style integrations with prescribed ocean conditions have been demonstrated (Kochkov et al., 2024; Baxter et al., 2025). Similar full-model emulators are being developed for climate application (Watt-Meyer et al., 2023).

The prospect is attractive for palaeoclimate science in particular. Transient simulations across millennia remain among the most expensive experiments in climate modelling, and a trustworthy fast atmosphere would transform what can be attempted. The obstacle is that the learned components of such models are fitted to a narrow slice of climate history. NeuralGCM was trained on reanalysis data of 2001–2018 (Kochkov et al., 2024), so any palaeoclimate

application is extrapolation by construction. However, the extrapolation behaviour of hybrid models has not been evaluated systematically, and a single-epoch test cannot distinguish which parts of the model fail, how quickly, and why. Existing evaluations concern the modern climate only (Baxter et al., 2025; Chen et al., 2026).

The Holocene offers a natural experimental design for this question. Orbital forcing is known precisely and varies smoothly, so a sequence of time slices between the present and the mid-Holocene forms a calibrated ladder along which the distance from the training climate grows step by step (Berger, 1978; Otto-Bliesner et al., 2017). The mid-Holocene itself is a benchmark of coordinated palaeoclimate modelling, with a well documented circum-global response pattern dominated by the amplified boreal summer insolation and the associated monsoon changes (Brierley et al., 2020).

In the present study we evaluate NeuralGCM along such a ladder. Six time slices at 0, 1, 2, 3, 4 and 6 ka are simulated with ten-member ensembles, forced by sea surface conditions taken from an 8300-year transient simulation of the coupled model AWIESM2, which simultaneously provides the verification target. A present-day baseline against ERA5 is established first, the response gradient is then examined component by component, and the learned precipitation scheme is tested against the moisture budget of the model’s own dynamics. Numerical stability is reported as part of the evaluation record, since the survival time of long integrations is observed to shorten as the forcing departs from the training climate.

2 Models and data

2.1 NeuralGCM

NeuralGCM combines a spectral dynamical core solving the moist primitive equations with neural-network parameterisations of unresolved physics, trained end-to-end on ERA5 (Kochkov et al., 2024). The deterministic 2.8° configuration is used here, with 32 vertical levels and learned physics acting every model hour. The released checkpoint is integrated through the published `PressureLevelModel` interface. Sea surface temperature and sea-ice concentration are prescribed as forcings, the atmospheric state is encoded once from an initial condition, and the model is advanced in daily steps. The stochastic precipitation variant of the model (Yuval et al., 2025) is used only in Sect. 6, since the deterministic checkpoint does not output precipitation.

One modification to the released checkpoint proved essential. Long integrations with the public weights fail systematically, at 200–300 days for the stochastic variant and during the second simulated year for the deterministic one, through a slow drift of global-mean surface pressure. The published remedy pins the global mean of log surface pressure at every dynamical step and was applied to all simulations in this study; the long integrations of Kochkov et al. (2024) and Baxter et al. (2025) rely on the same correction, which is however not part of the released weights. The consequences of this configuration detail are quantified in Sect. 7.

2.2 AWIESM2 and the Holocene transient simulation

The forcing and the verification target are taken from a transient simulation of the coupled model AWIESM2. The atmosphere component is ECHAM6 (Stevens et al., 2013) at T63L47 resolution, about 1.9° with 47 vertical levels, including the land surface scheme JSBACH. The ocean and sea-ice component is FESOM2, a finite-volume model on an unstructured triangular mesh with local refinement (Danilov et al., 2017). Model descriptions and mid-Holocene applications of this system are given by Shi et al. (2022) and Shi et al. (2023).

The transient simulation integrates the coupled system over the past 8300 years under time-varying orbital forcing, from 8.3 ka to 1950 CE. Monthly mean sea surface temperature

Table 1: Time-slice parameters. Orbital values follow Berger (1978) with the 0 and 6 ka anchors of the PMIP4 protocol. Insolation is the June-mean anomaly at 60°N relative to 0 ka; SST anomalies are 20-year means of the AWIESM2 transient relative to its own present.

Slice	Obliquity (°)	Eccentricity	Perihelion (day of year)	Insolation (W m ⁻²)	Global SST (K)	Tropical SST (K)
0 ka	23.450	0.0168	3	–	–	–
1 ka	23.631	0.0172	347	+1.1	–0.27	–0.31
2 ka	23.799	0.0176	330	+4.0	–0.30	–0.37
3 ka	23.958	0.0180	312	+8.2	–0.39	–0.43
4 ka	24.105	0.0183	295	+13.8	–0.36	–0.41
6 ka	24.105	0.0187	263	+23.3	–0.44	–0.54

and sea-ice concentration are available for every simulated year, which is essential here because the orbital signal expresses itself through the seasonal cycle. For each time slice a 20-year window of the transient centred on the target date is used, and the final decades of the run, corresponding to 1931–1950 CE, define the transient’s own present-day state.

2.3 ERA5

ERA5 (Hersbach et al., 2020) provides the initial condition sample, the present-day verification fields, and the base state of the boundary forcing. Monthly mean sea surface temperature and sea-ice concentration of 1990–1999 supply interannually varying base years, and pressure-level temperature and winds of the same period serve as the observational reference for the baseline evaluation.

3 Experimental design

3.1 Anomaly forcing and orbital parameters

Prescribing absolute sea surface conditions from a coupled model to a learned atmosphere fails in practice. The mean-state bias of the coupled model, which is inconsequential for a physical model, displaces the network from its training distribution and was found to trigger chronic warm drift. All palaeo forcings are therefore constructed by the anomaly method familiar from AMIP-style experimentation (Gates, 1992). The monthly climatology of a given slice, differenced against the transient’s own present-day window, is added to an ERA5 base state. Differencing within a single simulation removes the mean bias of AWIESM2 from the forcing, and anchoring the base state to ERA5 keeps the network inside its comfort zone while the palaeoclimate signal enters as a perturbation.

Orbital parameters are set per slice following the astronomical solution of Berger (1978), with the 0 ka and 6 ka values matching the PMIP4 protocol (Otto-Bliesner et al., 2017). Obliquity, eccentricity and the perihelion date are imposed in the solar geometry of the dynamical core. The resulting June insolation anomaly at 60°N rises monotonically from +1.1 W m⁻² at 1 ka to +23.3 W m⁻² at 6 ka (Fig. 1), while the prescribed global-mean SST anomaly deepens to –0.44 K. These two curves define the extrapolation-depth axis of the study. Greenhouse gas concentrations are not adjustable in the model version used, an unavoidable limitation discussed in Sect. 8.

3.2 Time-slice segment ensembles

Each ensemble member is an independent 14-month integration started from the same ERA5 initial condition. The first two months are discarded as spin-up, leaving one complete boreal

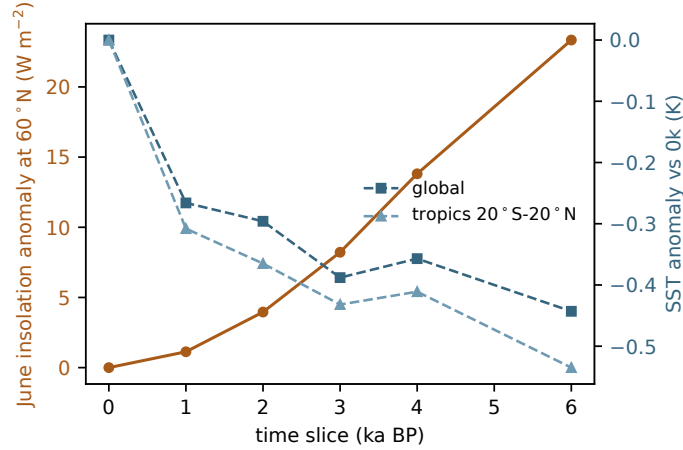


Figure 1: Imposed forcing gradient along the Holocene ladder. June-mean insolation anomaly at 60°N relative to 0 ka (orange, left axis) and prescribed SST anomaly of the AWIESM2 transient relative to its own present (blue, right axis) for the global mean and the tropics. Units: W m^{-2} and kelvin (K).

summer and one complete winter. The segment length is set from both sides, since a full December-to-February season requires the integration to cross into a second calendar year while the stability window of the weights favours short runs. Ten members per slice are generated, in the main protocol by assigning each member a different ERA5 base year of 1990–1999.

Running segments in parallel rather than as a restart chain is legitimate for this model class and was verified explicitly. The prognostic state contains no slow reservoir, as soil moisture, snow and ocean are absent or prescribed, and members started from identical initial conditions decorrelate within the first simulated month. The pairwise correlation of member anomalies reaches $-1/(N - 1)$, the algebraic expectation for fully independent draws, before the discarded spin-up ends. A caveat specific to seed ensembles was identified along the way. The deterministic encoder ignores its random seed entirely, so ensemble generation must perturb the initial conditions instead; a perturbation of 1 mK on the input temperature amplifies to weather-scale spread within one month and is used in the sensitivity protocol of Sect. 5.4. One representativeness check is reported for transparency. Sub-ensembles of four segments and of three continuous-run years correlate at only 0.29 in summer and 0.10 in winter when their 6 ka responses are compared, which reflects the small size of each sub-ensemble rather than a systematic difference, since merging them leaves the skill against the target unchanged.

3.3 Verification

Skill is always measured against the response of the transient simulation itself, computed from the same 20-year windows that generated the forcing. Forcing and answer key therefore originate from one physical realisation of the Holocene, and the pattern correlation measures whether NeuralGCM converts a prescribed ocean and orbit into the atmospheric response that the donor model produced, rather than whether two unrelated models agree. Pattern correlations are computed on anomalies from the area-weighted global mean. The forced signal is compared throughout against the ensemble noise, defined from the member spread of the 0 ka reference, since interpretation of a correlation is impossible without knowing whether a predictable signal exists at all (Deser et al., 2012).

The target itself is a finite sample and imposes a ceiling on attainable skill. Splitting each 20-year target window into decadal halves and correlating the two half-window responses yields, after Spearman–Brown adjustment, the correlation a perfect atmosphere could reach against

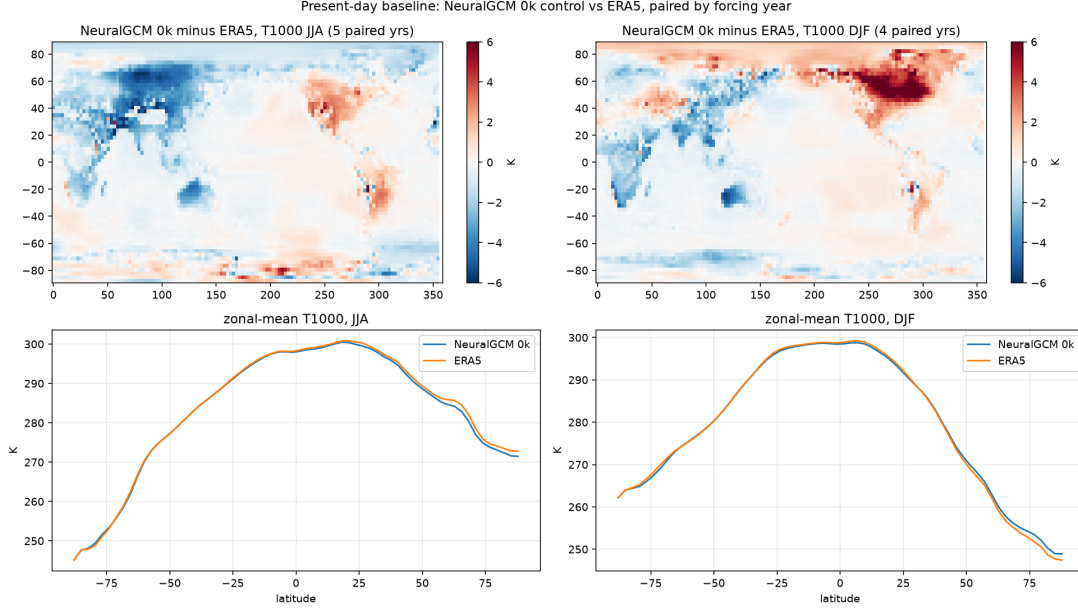


Figure 2: Present-day evaluation of the 0ka control against ERA5, paired by forcing year. Bias of 1000 hPa temperature (K) for JJA and DJF (top) and zonal-mean temperature of the control and ERA5 (bottom).

this target. The ceiling ranges from 0.62 to 0.93 depending on slice and season. Reported correlations are quoted in absolute terms with this ceiling alongside as a reference, rather than as normalised ratios, since the ceiling itself rests on only two independent half-windows and carries a wide uncertainty.

4 Present-day baseline

A difference experiment cannot expose a bias shared by both epochs, so the control climate is evaluated on its own terms first. The 0ka configuration was integrated continuously for eleven years under observed monthly SST of the 1990s and compared against ERA5 pairwise by forcing year, so that both sides contain the same El Niño events and the same Pinatubo cooling and interannual variability largely cancels.

The control reproduces the modern atmosphere closely (Fig. 2). The global-mean temperature bias at 1000 hPa is -0.38 K in boreal summer and -0.04 K in winter, with a root-mean-square error near 1.3 K and a pattern correlation of 0.997 against ERA5; the 850 hPa zonal wind correlates at 0.971. The bias does not grow with integration time, as the annual global offset fluctuates between -0.04 and -0.56 K across ten years with no trend, which establishes that integration length is not a limiting factor for this study.

The error that does exist has a definite address. Ocean regions sit within a quarter degree of ERA5 while the summer bias reaches -1.08 K over the Sahara and Sahel belt and -1.35 K over Europe. The weakness of the model concentrates over land, where its physics must stand in for an interactive surface it does not possess. This pattern recurs in the palaeoclimate results below.

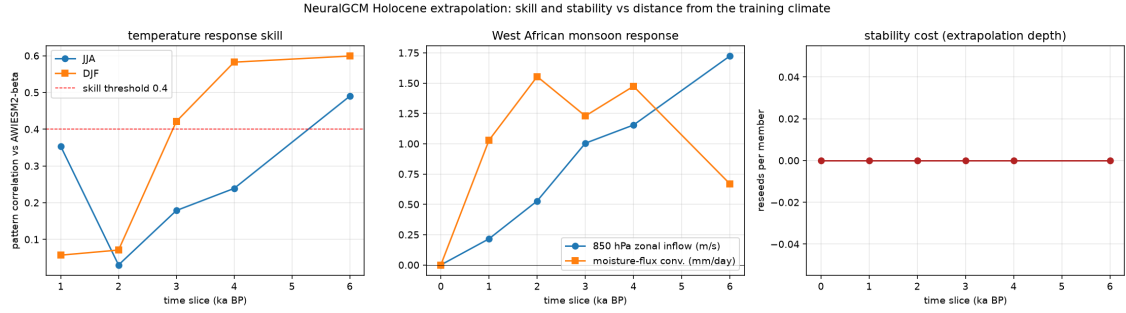


Figure 3: Response and skill along the Holocene ladder from ten-member ensembles. Pattern correlation of the near-surface temperature response against the AWIESM2 transient for JJA and DJF (left), West African monsoon circulation response (middle), and stability cost (right), as a function of time slice. The 0 ka reference has no defined correlation and is omitted from the skill panels.

5 The Holocene response gradient

5.1 Circulation follows the orbit

The dynamical response tracks the forcing ladder step by step (Fig. 3). The 850 hPa westerly inflow into West Africa, the classical dynamical signature of a strengthening African monsoon, increases monotonically through +0.22, +0.53, +1.00, +1.15 and +1.72 m s^{-1} from 1 ka to 6 ka, roughly proportional to the insolation anomaly. No slice shows a reversal or a plateau. The vertically integrated moisture-flux convergence over the monsoon region responds with the same sign at every slice. Whatever happens to the learned physics at depth, the resolved circulation keeps converting the imposed insolation and SST gradients into the expected dynamical response across the full range tested.

5.2 Winter skill rises with the signal

The winter temperature response is captured increasingly well as the ladder descends, with pattern correlations of 0.06, 0.07, 0.42, 0.58 and 0.60 from 1 ka to 6 ka. Skill improving with extrapolation depth appears paradoxical only until the signal size is considered. At 1 and 2 ka the imposed insolation anomaly is a few W m^{-2} and the target response of the donor model amounts to a few tenths of a kelvin, so there is almost nothing to predict. The signal-to-noise ratio of the forced response rises from 1.1 at the shallow end to 2.0 at 6 ka, in step with the correlation. The low shallow-end correlations measure the weakness of the forced signal relative to internal variability, not a failure of the model. At 6 ka the achieved correlation of 0.60 stands against an attainable ceiling of 0.74, so the winter response is captured nearly as well as the finite target allows. The global winter cooling at 6 ka reaches -0.73 K against -0.83 K in the target, indicating that the thermodynamic response to the prescribed ocean is essentially correct once it is large enough to detect.

5.3 Summer stays weak wherever it is measured

The summer response never reaches comparable skill. Its pattern correlation remains below 0.5 at every slice and does not order itself with depth, even where the signal-to-noise ratio matches that of the successful winter cases. Summer is the season in which the land surface dominates the mid-latitude and monsoon response, through soil moisture, vegetation and surface-flux feedbacks that the model represents only implicitly through statistics learned from the modern climate. The same fingerprint appeared already in the present-day control as a summer cold

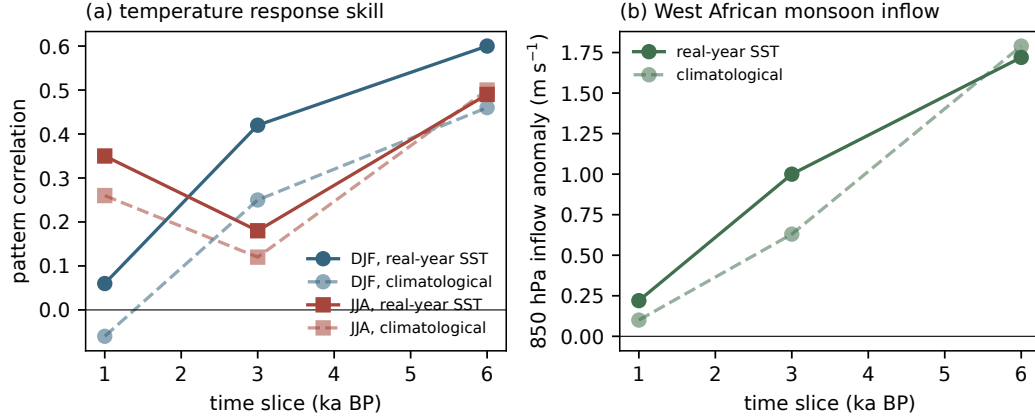


Figure 4: Sensitivity of the gradient results to the forcing protocol. (a) Pattern correlation of the temperature response for the main protocol with real base years (solid) and the climatological protocol with perturbed initial conditions (dashed). (b) Same as (a) but for the 850 hPa West African inflow anomaly (m s^{-1}).

bias over land. The ceiling analysis sharpens the contrast, since the summer target at 6 ka is the most reproducible of all cases with a ceiling of 0.93, yet the achieved correlation reaches only 0.49, while winter attains 0.60 of an attainable 0.74. The consistency of this weakness across the baseline, the gradient and the precipitation test below points to a structural limitation rather than a sampling artefact, and enlarging the ensemble does not remove it.

5.4 Insensitivity to the forcing protocol

The findings above survive a deliberate change of experimental protocol (Fig. 4). A second set of ensembles replaces the interannually varying base years by the ERA5 monthly climatology, widens the anomaly windows from 20 to 100 years, and generates spread purely by perturbed initial conditions. Every headline feature replicates. The monsoon inflow remains monotonic and reaches $+1.79 \text{ m s}^{-1}$ at 6 ka against $+1.72$ in the main protocol, winter correlations again rise with depth, and summer correlations again stay low and unordered. Amplitudes are slightly weaker under the smoother forcing, as expected. The component-wise split therefore does not depend on how the boundary condition was built.

6 Precipitation as the failure mode

Precipitation requires the stochastic variant of the model, and its response separates cleanly into a dynamical part that works and a learned part that does not (Fig. 5). In a five-member ensemble at 6 ka the rainfall response over the West African monsoon region is $-1.06 \pm 0.36 \text{ mm day}^{-1}$, opposite in sign to the $+2.68 \text{ mm day}^{-1}$ wetting of the target model, with four of five members negative. The simulated rain belt does not shift northward, whereas the target shifts by about two degrees.

Everything upstream of the rain is nevertheless correct. The 850 hPa inflow strengthens by $+0.70 \text{ m s}^{-1}$, and the vertically integrated moisture-flux convergence computed from the model’s own winds and humidity yields $+1.65 \text{ mm day}^{-1}$ over the same region, matching the target’s $+2.61 \text{ mm day}^{-1}$ in sign and magnitude. The dynamics deliver the moisture and demand the rain; the learned precipitation operator, fitted to the relation between circulation and rainfall in the modern climate, declines to produce it under an orbital configuration it has never seen. This is the split the study was designed to detect, caught within a single model

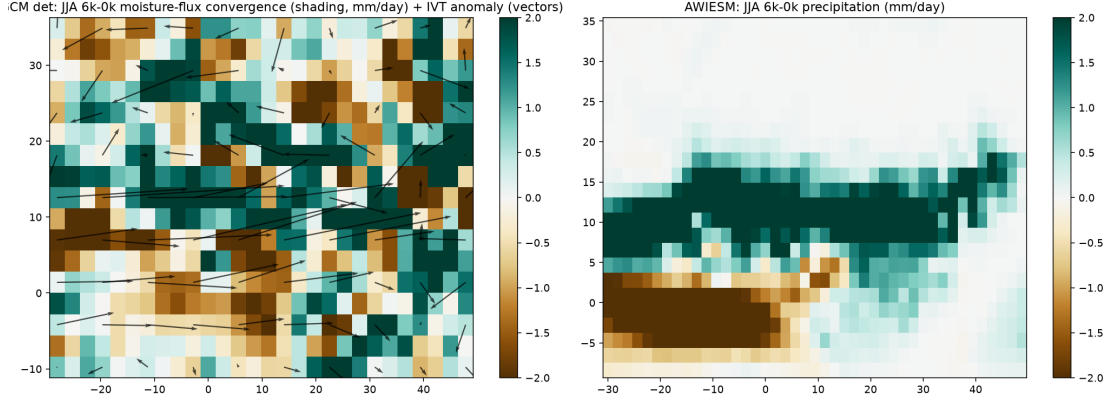


Figure 5: Mechanism of the monsoon precipitation failure at 6 ka. Moisture-flux convergence response of the deterministic model (shading, mm day^{-1}) with the integrated vapour transport anomaly (vectors) over Africa (left), and the precipitation response of the target model AW-IESM2 (right). The learned precipitation response (not shown) is opposite in sign to both.

in a single variable. A practical consequence follows immediately. Palaeo-monsoon work with this model class must run through dynamical diagnostics such as moisture convergence, and any application resting on the learned rainfall output is presently unsupported. A slow erosion of simulated wet extremes in long integrations has been reported independently by users of the same released model, consistent with this picture.

7 Stability of long integrations

Numerical survival itself carries information about extrapolation depth (Fig. 6). With the released weights every long integration fails, through upper-level blow-up at 200–300 days for the stochastic variant and a warm runaway during the second year for the deterministic one, across all seeds, ocean forcings and hardware tested. Pinning the global-mean log surface pressure removes the deterministic failure entirely at the present-day end, where an eleven-year integration completed without incident, but only postpones failure at depth, where the 6 ka configuration reliably terminates near day 1230. Survival time thus shortens as the forcing departs from the training climate, although this rests on one long integration per endpoint and is reported as an observation rather than a calibrated measure. The production ensembles of this study, one hundred segments of 425 days, completed without a single intervention, which places the working configuration comfortably inside the stable regime. Failure statistics of this kind are reported here deliberately, since discarded members are part of the evaluation record of a learned model rather than a technical nuisance.

8 Discussion

The central result is that the palaeoclimate limit of this hybrid model is not a date but a decomposition. Nothing breaks abruptly along the ladder between the present and 6 ka. The resolved dynamics extrapolate to the deepest slice tested without measurable degradation, the thermodynamic winter response follows the prescribed ocean once the forced signal exceeds internal variability, and the land-mediated summer response is unreliable at every depth, culminating in a sign error of the learned monsoon rainfall. For applications this translates into a usability map. Circulation and moisture-transport studies are defensible to at least 6 ka, winter temperature reconstructions are defensible where the signal is strong, and any conclusion

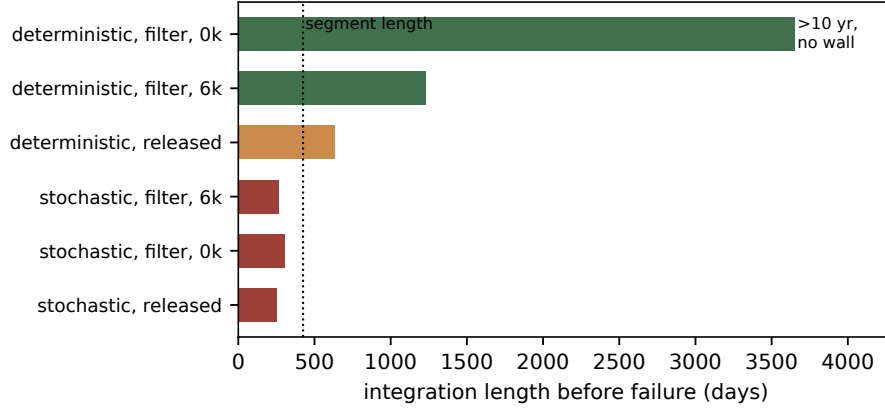


Figure 6: Integration length before failure for the released checkpoints and for the corrected configuration at 0 ka and 6 ka. The dotted line marks the 425-day segment length used in the production ensembles. The 0 ka corrected configuration shows no failure within the eleven years tested.

resting on learned precipitation or on the summer response over land is not.

The mechanism behind the map is the architecture itself. The primitive equations know nothing of the training period, so the circulation response to a modified orbit is genuine physics and travels well. The learned components encode the covariances of 2001–2018, and the mid-Holocene breaks precisely those covariances, most severely where the land surface couples radiation, hydrology and rainfall in ways the training data cannot constrain. The failure of the rainfall operator despite a correct moisture budget makes this concrete, since both quantities live in the same simulation and differ only in how their physics is represented. These results suggest, as a hypothesis to be tested on other systems, that the extrapolation fidelity of a component of a hybrid model scales with the share of its physics solved by the dynamical core rather than fitted to data. The winter success deserves one qualification in this reading, since it demonstrates that the prescribed ocean carries the forced signal through largely resolved thermodynamics, not that the learned components themselves extrapolate. Survival time of the numerical integration also shortens with extrapolation depth, an observation consistent with this picture though based on few long runs.

Several limitations bound these conclusions. Greenhouse gas concentrations cannot be specified in the model version used, so the small mid-Holocene CO_2 deficit enters only indirectly through the prescribed ocean, and epochs with large greenhouse departures are outside the scope of this design. Monthly means are compared on the fixed modern calendar, and the known calendar effect of orbital precession (Joussaume and Braconnot, 1997; Bartlein and Shafer, 2019) shifts monthly diagnostics by amounts that are similar for both models being compared but not zero, and is left unquantified here. The verification target is a single coupled model, and the mid-Holocene monsoon is a region of substantial inter-model spread (Brierley et al., 2020), so the pattern correlations reported here measure fidelity to one plausible Holocene, not to the real one. The anomaly method imposes the change of boundary conditions rather than their absolute palaeo state. Finally, all results concern one hybrid model at one resolution, and the decomposition found here invites testing on other systems of the same class (Watt-Meyer et al., 2023).

The design itself may be the most transferable contribution. A forcing ladder anchored in known orbital mechanics, an anomaly method that keeps a learned model inside its distribution while feeding it a palaeoclimate signal, verification against the donor simulation that generated the forcing, and stability accounting as an independent depth gauge together form a portable

protocol for auditing any fast learned atmosphere before it is trusted with the past.

9 Conclusions

NeuralGCM was evaluated along six Holocene time slices with more than one hundred independent ensemble segments, against a present-day baseline from ERA5 and a palaeoclimate target provided by the AWIESM2 transient simulation that also supplied the forcing. The present-day control is accurate to a few tenths of a kelvin globally, with its error concentrated over land. Along the ladder the model separates into a dynamical component that follows the orbital forcing monotonically to 6 ka, a winter thermodynamic component that is skilful wherever the forced signal is detectable, and a land-dominated summer component, including the learned rainfall, that fails independently of extrapolation depth. Skill is limited by what the parameterisations were able to learn from eighteen years of the modern climate, not by how far back the model is pushed. These results define the terms under which hybrid models can already serve palaeoclimate research, and the terms under which they cannot yet.

Code and data availability. NeuralGCM and its released checkpoints are publicly available. Simulation output, forcing files and analysis scripts of this study are archived on the DKRZ Levante system and available from the author on request.

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